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RESEARCH-ARTICLE

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NewsMead: A user-centric design of a news personalization application for older adults

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Abstract

The transition from traditional print to online platforms for newspapers has presented challenges for older adults, who may struggle with unfamiliar digital interfaces. However, as older adults increasingly engage with digital content, there is a growing need for assistive technologies to support their digital literacy and inclusion. This study addresses this need by developing a user-centric news personalization application tailored to the needs and preferences of older adults. The initial application prototype was developed based on a literature review, prototyping, and wireframing. The results revealed that older adults do not actively consume news due to irrelevant content and perceive the activity outside their lifestyle. When provided with a news application considering their difficulties, they fostered a renewed interest in staying informed and connected with current events.

CCS Concepts

• Human-centered computing → User interface design; • Information systems → Recommender systems.

Keywords

Older Adults, News Recommender Systems, User-Centered Design

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1 Introduction

Over the years, older adults have encountered challenges in staying informed about recent events due to the changing landscape of news consumption. While traditional newspapers have declined, online platforms have become increasingly dominant [10]. However, older adults are slowly becoming heavy consumers of digital content, using social media platforms like Facebook and YouTube. They are challenged by complicated and differing interfaces across these news platforms [13]. Specifically, they find it difficult to browse broad menus, read small font sizes and icon labels, and navigate between interfaces. In addition, some are physically frail and possess some cognitive conditions caused by aging [9]. In response to these challenges, there is a growing need for assistive technologies and user-centered design approaches to support older adults in their digital news consumption habits.

Older adults also face issues interacting with applications [14]. The use of standardized icons and symbols often fails to accommodate the diverse cognitive abilities and preferences of older adults, creating usability barriers for this demographic when interacting with digital devices [12]. Moreover, a study [16] compared older adults' performance on interfaces with varying cognitive loads, revealing the significant impact of cognitive demand on task success rates. Tasks characterized by high time pressure yielded inconsistent outcomes among older adults, underscoring the complexity of cognitive factors contributing to their heterogeneous user experiences.

Additionally, older adults tend to forget where they gather information sources online. Older adults commonly consume social media content and are therefore exposed to fake news and unreliable news sources [3, 4]. This prevents them from sharing information because they are unsure of its reliability and value [5]. A news personalization application will empower older adults to read and share news safely and directly from reliable news sites.

A news aggregator combines sources without considering users' individual preferences and interests [2]. While it aggregates news from various publishers into a single feed for convenience, this approach may not effectively cater to older adults' diverse needs and preferences. In contrast, recommendation algorithms, commonly used in news applications today, offer a more personalized

experience by analyzing user interactions and learning their preferences over time [11]. By incorporating recommender systems, the application can better understand the specific interests of older adults and tailor the news feed accordingly, enhancing their overall reading experience [8]. This personalized approach acknowledges that older adults often learn and engage with content based on what they encounter rather than actively seeking information through manual searches [8].

A news personalization application explicitly tailored for older adults will revolutionize their news consumption experience by providing curated content from reliable news sources. The recommender engine learns and adapts to each individual's interests by analyzing user interactions and preferences, effectively personalizing their news feed. This personalized approach empowers older adults to stay informed in a way that suits their unique needs and interests.

2 Methodology

Through a user-centered design approach, the methodology involves a needfinding assessment, focus group discussions with usability testing, and user acceptance testing to gather insights into older adults' behaviors, preferences, and pain points when interacting with the application. The initial prototype was developed based on insights from a review of existing literature and best practices [6, 7, 15], ensuring that older adults' needs are considered.

In seeking participants, the researchers asked friends and relatives who are older adults or know others who may want to participate in the study. Upon receiving their contact information, the researchers informed them about their role in the study. All participants are physically and mentally able to use mobile phones by themselves. Participants were asked to bring Android phones. If they have none, the researchers provided it.

In conducting the needfinding and focus group discussions, the researchers and participants agreed upon the schedule and venue for the focus group discussion. A total of 4 community-dwelling Filipino older adults were purposefully selected. The small number of participants is based on the study's objectives and consistent with the recommended sample size in health-related inquiries that ensure active participation of all respondents [18]. Furthermore, it ensures that the discussion is manageable and that participants can freely discuss their needs, perceptions, and opinions. The needfinding was conducted first, followed by three iterations of focus group discussions spread out on separate dates.

The needfinding was conducted to gain insights into the design needs of older adults for smartphone user interfaces. This phase elicits their preferences, needs, and opinions about smartphone user interfaces and news-accessing habits. The assessment phase commenced through an FGD. This included a usability testing session where participants interacted with the app. During this session, participants were asked to perform specific tasks using the app, which allowed them to experience its features and identify any usability issues. Following the hands-on testing, participants discussed their experiences, providing feedback on the app's interface, functionality, and overall user experience. The data collected was subjected to qualitative analysis. We manually transcribed the audio because the language used was a mix of Filipino and English.

Thematic analysis was conducted on the audio to extract insights which helped us gain deeper insights into the app's usability from the perspective of older adults. This method was preferred over a questionnaire because older adults may find it challenging to engage with objective-type questions or may struggle to articulate their thoughts in written form fully. Additionally, thematic analysis allows for capturing nuanced feedback that might otherwise be lost in translation, ensuring a more accurate and comprehensive understanding of their experiences and needs. All the sentences referenced in the thematic analysis are translated into English. The assessment findings are then used in the subsequent design phase. They inform the development of a prototype of the smartphone user interface tailored to the unique preferences and requirements of older adults, contributing to the overall improvement of their user experience. Once the needs assessment of the target users is conducted, the requirements will update and reflect their specific needs.

After improving the application from the 3 FGDs, seven (7) participants were then interviewed for UAT, excluding the participants during the FGDs to avoid bias. A set of open-ended questions based on the Post-Study System Usability Questionnaire (PSSUQ) were asked during each interview, and the responses were transcribed by the researchers [1]. Statements were extracted from each interview and coded with initial categories resulting in 33 categories. The categories were revised in a feedback loop by merging similar categories to refine and reduce them to main categories. The main categories were based on the sub-scales of the PSSUQ, which were System Usefulness, Information Quality, and Information Quality. However, System Usefulness was split into two categories, Perceived Usefulness and Perceived Ease of Use, for a clearer distinction. Categories that do not fall under any similar category were converted to their standalone categories.

2.1 User-Centered Design Approach

The UI and UX design for NewsMead employs a User-Centered Design approach for its design process. The method involves iterating between identifying 4 phases: the context of use, design requirements, design solutions, and design evaluation.

2.1.1 Context of Use. The focus group discussion targets older adult users aged 60 years and over with existing cognitive, perceptual, or physical decline brought by aging, all residing in the Philippines. This is to involve elderly users early in development to understand their specific needs and produce accurate solutions. The intended use of the application NewsMead is to provide an elderly-friendly platform for accessing news content through a news personalization application. The users must also be familiar with using an Android smartphone to operate the application. Users engaged with NewsMead through usability testing, during which they were assigned tasks to navigate the application, browse articles, and save them. Given potential challenges such as cognitive and physical decline or limited technological familiarity, the interface will adhere to design principles and guidelines from existing literature tailored to older adults to fit their needs.

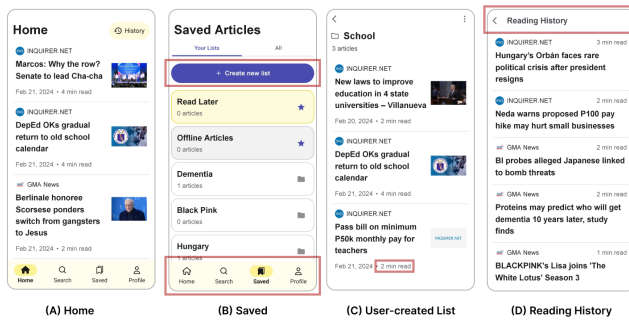


Figure 1: NewsMead Interface Designs for the Home, Reading History, Saved and user-created lists

2.1.2 Design Requirements. To produce the desired design solution, the design requirements are first identified. The requirements consider visual, interactive, and cognitive elements suitable for older adults. Many of the specified requirements and design considerations were based on the literature review recommendations and reviewing existing design guidelines and considerations for older adults from literature by Sara J. Czaja, Walter R. Boot, Neil Charness, Wendy A. Rogers [15], Chirayus and Nanthaamornphong [7], and Cheng and Sabran [6]. These requirements focus on visual, interactive, and cognitive elements to ensure a user-friendly experience. Key considerations include using elderly-friendly fonts such as Arial, larger font sizes (minimum size 12), and customization options. Iconography should employ task-relevant, skeuomorphic designs with text labels to aid understanding. High contrast and simple color combinations are essential to reduce visual burden. The interface design should be flat, minimal, and aesthetically pleasing, while the layout should simplify tasks with evenly distributed visual elements, prioritizing frequent operations. Button properties require large buttons with adequate spacing, avoiding placements at the bottom or left. Input methods should include shortcuts, graphical passwords, and simplified decision-making processes. Presentations should be simple and consistent to manage cognitive load, with minimized irrelevant details and reduced procedural steps. Guidance materials and search functionalities should support navigation, while error handling should offer gesture alternatives, multiple warnings for critical actions, and “undo” options.

2.1.3 Design Solutions. This phase of the design process involves producing a design solution by applying the specified requirements that meet the user’s needs. The design solution focuses on creating an accessible user experience for older adults, constantly involving them by asking for feedback through focus group discussions and observations during usability testing for each iteration. The study’s design solution phase involved three (3) iterations of the prototype, including designing, development, and testing, before moving to the next iteration. The iterative nature of this phase allows for continuous refinement, ensuring that the final design aligns seamlessly with the intended purpose of the news personalization application, NewsMead, for the elderly.

The interface designs were produced through the identified requirements, as seen in Figure 1 and Figure 2, which shows the

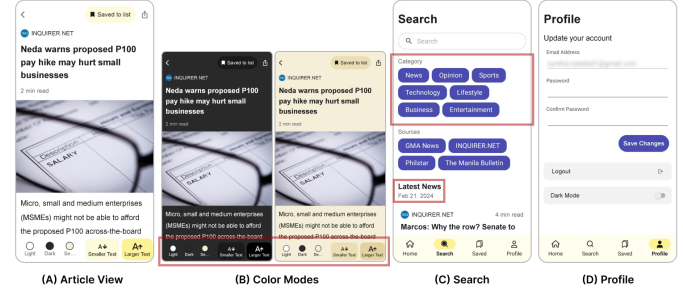


Figure 2: NewsMead Interface Designs for the Article view, Color modes, Search, and Profile

application’s current designs. Figure 1 shows consideration for the requirements established under *Design Requirements*, specifically iconography, interface design, and the cognitive load and attention of the users for ease of use and simple navigation. The navigation design is kept simple, and the application utilizes a modern flat design to reduce information density and complex patterns, as seen in Figure 1 (a). The icons convey purpose and function with accompanying icon labels for easy identification for the users highlighted in Figure 1 (b). The estimated length of reading minutes is also presented to provide a solution to narrow down the choices for what the users can read based on their speed preference or timeliness, seen in Figure 1 (c). Furthermore, the button properties and error handling were also considered when designing for actions that elderly users will take. In Figure 1 (b), the design solution opted for larger button sizes, and essential buttons are placed at the top of the screen to be easier to find. The app also includes keeping the user’s reading history, in Figure 1 (d), to provide backtracking and allow users to return to previous articles.

As for the user’s reading experience designs, the user’s preferences were also considered in Figure 2. The requirements from Table 1 for typography, color and contrast, layout, and hierarchy were accounted for to allow users to have a customized reading experience that accommodates their needs. The font family *Arial* is being used as an elderly-friendly font throughout the application, and the font sizes used a majority of 20px with a minimum of 14px, evident in texts seen in Figure 2 (a). The colors used throughout the app used fewer color variations to provide a solution for reducing the visual burden of older adults and provide a high color contrast through the use of dark colors on top of a light background. The application applies reduced color intensity without over-saturation or distracting illustrations or advertisements. Figure 2 (b) presents the option for multiple color modes to suit their reading preferences. Customization functionalities for reading articles are placed at the bottom of the screen to provide quick access buttons, also highlighted in Figure 2 (b). Aside from these, the users’ input method and cognitive limitations were considered, designing procedures requiring fewer paths to help with cognitive burden and make the tasks easier to complete. This is shown through Figure 2 (c) with the ability to search articles easily by category or source and having the latest news on the search page for simpler navigation. The users

Table 1: Application Feature List

Function	Description
Register/Log-in	Account creation and management.
View Article	Read articles within the app.
Customize reading	Adjust font, contrast, and color mode.
Search/Filter	Find articles by keyword, topic, date, category, or source.
Bookmark	Save articles for later access and organize them into collections.
Read Later	Save articles for future reading.
Offline reading	Download articles for reading without internet access.
Share articles	Share articles on social media or messaging apps.

can also update information or log out of their accounts without additional steps in the Profile, shown in Figure 2 (d). These aim to provide users with better efficiency, processing, and memory.

2.1.4 Design Evaluation. The design solution and prototype were evaluated through focus group discussions and usability tests. Usability tests were conducted to observe and analyze how the users navigate the app by performing tasks relevant to testing the user's experience and interaction with the application. The group discussions were conducted to gain feedback after the usability testing to understand improvement points for succeeding prototype development.

2.2 Application Features

The application, NewsMead, is a news personalization app for older adults that will contain several relevant features for basic application interaction and for reading and interacting with news articles and content. The application contains four (4) main screens that the users navigate between. These are Home, Search, Saved, and Profile. *Home* contains the user's news feed, showcasing recent news. *Search* contains the functions for users to search and filter the articles. *Saved* houses the saved and bookmarked articles of the users and collections that the users created. Lastly, *Profile* contains information about the user's profile and serves as a place to edit their information or log out of the application. Table 1 details the functions of each initial application feature.

3 Results

Discussions were audio-recorded with the participant's consent. The focus group discussion (FGD) began with the researchers introducing themselves and explaining the ethical considerations. After having the participants sign the consent forms, the discussion formally began. One researcher facilitated the FGD, while others assisted as needed.

The first FGD began with a needfinding assessment comprising questions to understand the participants' experiences and behaviors related to news consumption. These questions covered topics, including the participants' daily smartphone usage, online access

challenges, preferred methods for finding and reading news articles, and experiences with existing news providers. Additionally, participants were asked about any limitations or health issues that may impact their ability to access news content. After completing the needfinding assessment, the participants were provided a five-minute break to transition into the first usability testing phase. During usability testing, one researcher demonstrated the application to the older adults, guiding them through its features and functionalities. Meanwhile, the other researchers observed the participants' interactions with the app, taking note of their habits, preferences, and any challenges encountered. They also provided assistance as needed. After the demonstration, the participants were given ten (10) minutes to freely browse the application.

We then discussed with the participants to gather feedback on their experiences using NewsMead. Participants were encouraged to share any improvements or suggestions they had, as well as their overall impressions of the app. A researcher facilitated this by asking open-ended questions and prompting participants to elaborate on specific features or functionalities they found useful or challenging. This dialogue provided valuable insights into the usability and effectiveness of the NewsMead from the perspective of older adults. This is repeated for succeeding iterations.

3.1 Needs Assessment

The thematic analysis for the needs assessment uncovered five (5) key themes: (1) Smartphone experience; (2) Irrelevant content; (3) News accessing habits; (4) High interest topics; and (5) Reading Perceptions.

3.1.1 Smartphone Experience. The participants had varying experiences when it came to using their smartphones. All admitted not being good with technology and that they tend to rely on others for help in technology, such as P4. However, while they are not knowledgeable about gadgets, they can use smartphones mainly for communication, including entertainment and e-commerce, as P1. When it comes to reading news, only P2 actively uses smartphones for news while the others only use it as an alternative for accessing news, such as P4. The participants would mostly use Google or YouTube if forced to use their smartphones to find news and were unaware of any existing news applications.

"I am not good with gadgets. He [spouse] is the one who feeds me news. He is the one who knows how to use gadgets" - P4

"I use it for calling and texting, and if I am on watch duty, I just watch [on smartphone] what is on" - P1

"I use the cellphone for checking Facebook. To check what the latest news is" - P2

"I usually prefer television, and then I will find news on my cellphone." - P4

3.1.2 Irrelevant content. A shared sentiment among the participants is that there tends to be repetitive content. They complained about current events and politics being repeated multiple times. P3 and P1 explained their frustrations, which all the participants strongly agreed with. It became frustrating for the participants to receive the same news repeatedly. Additionally, the participants expressed frustration with advertisements as well. P1 presented an example of their experience with advertisements. Due to the

repetitiveness of news, P4 went for an alternative, which is vloggers, because they are unsatisfied with the news they currently receive. However, P2 added that sometimes they are fake, which added to the challenges that older adults face due to the irrelevant content they receive when consuming news.

"You know, news has been too repetitive. First thing in the morning until tomorrow, it is all the same!" - P3

"Sometimes it is even saying the same news!" - P1

"What would happen is we would go to others, the vloggers" - P4

3.1.3 News accessing habits. The participants read news daily. They often read the news in the morning or when they have nothing to do. P2 includes reading the news in their daily routine. P4 adds that they read it during downtime but also said they, along with their spouse, can not read at night. In the case of P1, he explained that he only reads in the morning because someone gives him a newspaper. However, while the participants may include it in their daily routine, such as P3, they do not always read it due to a lack of interest. This sentiment is also common with other participants. While traditional article-based news consumption may decline due to waning interest, the participants actively seek out news through alternative mediums. They tend to favor watching it on television or YouTube or listening to it on the radio. When the participants were asked how they find news, all participants mentioned YouTube, Google, and television when looking for news. The participants enjoy accessing news in video format. However, when the participants were further asked if they also actively read articles, they did not, but instead, listened to the radio, which P3 and P4 emphasized. Only P2 reads news on social media. They mentioned Facebook as the social media they use to access news, which surprised the other participants because they were unaware that they could also read news on Facebook, an application they were all actively using.

"It is when I wake up in the morning, and when I do not have anything to do yet. It is cause I am always home" - P2

"But when we are in bed or about to fall asleep, we could not read anymore" - P4

"Because if you are not interested in the news, there is nothing you can do." - P3

3.1.4 High interest topics. It was observed that the participants actively engage with news regarding entertainment, current trends, and gossip. P4 personally does not consume news as often anymore. Instead, they read or listen to gossip or celebrities. It also led P3 to search for gossip because current news does not interest them. The other participants also agreed that they usually seek gossip as a news topic when reading articles; otherwise, they try to look up anything that is currently trending. In line with what was mentioned under *News accessing habits*, the participants turn to entertainment because they find news on current events too repetitive or uninteresting.

"Usually, celebrities are more popular today. Gossip, artists, that is what I usually read today" - P4

"We watch news about gossip instead so that we can have something new" - P3

3.1.5 Reading Perceptions. The participants had similar perceptions regarding reading news. All expressed aversion towards reading and prefer listening to the radio to reading the news. P4 summarized that reading becomes difficult in their lives as older adults and is not for them. The others agreed, stating that reading the news is not in their lifestyle. Only P1 actively reads the news, but only because someone gives him a newspaper in the morning. They had no interest in reading the news because they generally perceived it as a chore and that it was for students or businessmen.

"For news that are articles, we are no longer used to them. Because in our way of living, we are at a point where we are just—this. We watch whatever, but if you limit us to news articles, you need to learn it; it is like you are a student" - P4

In the morning, there is someone who always gives me a newspaper, so I read" - P1

3.2 Usability Testing

3.2.1 Iteration 1. Participants appreciated the overall user experience, especially the simplicity of the setup and the ability to adjust text size. They found navigation easy and enjoyed the features like color modes, reading history, and the search function. However, they desired more local and entertainment news and articles translated into the local language. Accessibility improvements, such as better device compatibility and clearer feature cues, were also suggested.

3.2.2 Iteration 2. New features like the "translate to Filipino" button and the "read-aloud" function were well-received. Users found the app easy to use and appreciated renaming the "filter" button to "simplify." Despite the improvements, participants still wanted more local news and experienced technical issues with the new features. They suggested including more relatable and localized content for the masses.

3.2.3 Iteration 3. Participants were satisfied with adding a Filipino-only news button and improved button placements. They found the app increasingly user-friendly and praised its simplicity and readability. The app was compared to a newspaper, and users appreciated its reliability and lack of advertisements. They emphasized the preference for local news and expressed that the app could replace social media for news consumption.

3.3 User Acceptance Testing

The UAT resulted in seven (7) main categories which are: (1) Perceived Usefulness, (2) Perceived Ease of Use, (3) Interface Quality, (4) Information Quality, (5) News Content, (6) Branding, (7) Technical Issues. Table 2 shows the assessment of the categories extracted from the statements of the participants. Relevance was computed using four (4) different values adopted from the paper of [17] which are: (-1) explicit rejection, (0) no reference, (1) reference, and (2) reference with an accent. Explicit rejection is a reference that negates the category. An example statement is: *"The save to list button is not that obvious. It is not visible, especially in color."* -P6 which negates the Perceived Ease of Use category. No reference is a statement that does not mention anything to the category while a reference does. A reference with an accent adds emphasis to the statement such as: *"It is very clear. Very good!"* -P5 where P5 adds emphasis to their

Table 2: Assessment of categories

Category	Relevance	Frequency	I-F
Perceived Usefulness	73	84	7
Perceived Ease of Use	84	72	7
Interface Quality	7	32	6
Information Quality	10	27	5
News Content	-8	9	4
Branding	-6	6	3
Technical Issues	-7	11	3

thought. Frequency is the number of references to the category and Interview-Frequency (I-F) is the number of interviewees that were able to reference the category.

4 Discussion

4.1 Needs Assessment

Older adults use smartphones mainly for communication and entertainment. Despite the lack of expertise in technology, they learned to use smartphones because they are essential today. However, they do not often use it for news. Instead, they turn to alternative mediums to stay informed. The participants often use YouTube or radio to listen to the news. This is because they experience difficulty reading the news. They already access it daily and often find time to read news, but they face challenges with it. Older adults encounter repetitive content and fake news. It gets difficult to distinguish fake news due to vision deterioration brought on by aging. These challenges contribute to a sense of disinterest among older adults, leading them to seek alternative mediums for news consumption or not consume news at all. To discover fresh and interesting news, some turn to vloggers or gossip to fulfill their desires. Some already only read about gossip or celebrities due to repeated dissatisfaction with current news. They do not find a need to stay informed anymore because they believe it is no longer part of their lifestyle. Reading news articles became a chore and was perceived to be only for students and businessmen. If they want to be informed, they would rather listen to the radio than read.

This presents a need for older adults to feel like they can read the news, too. They perceive reading the news as an academic activity and a challenging task. The presentation and algorithms of news outlets have made it difficult for older adults to enjoy news consumption, which presents a need for older adults to have a tailored algorithm and news sources to provide an easier and safer reading experience. Since some older adults face difficulties with vision deterioration or prefer listening to reading news, they may also prefer having an audio option for their news consumption.

4.2 Usability Testing

After three iterations, the application added a translation feature to accommodate the preferred language of other users. It also added a read-aloud feature because the participants often prefer listening to news articles. The filter button was renamed to simplify because a participant did not understand what filters do on the internet. Lastly, a button was added to switch to only Filipino articles to

provide local news the participants wished to see or focus on. The participants express the importance of local news because it is something closer to them and is for the masses. They consider themselves part of the masses and prefer to see news that is easier to understand. After completing the last iteration, the participants were satisfied and said it was already perfect for their use.

4.3 User Acceptance Testing

The UAT revealed that older adults found NewsMead to be a useful and intuitive news personalization application. Participants appreciated features such as the search function and text resizing, which enhanced their reading experience by making news from various local sources more accessible. Users praised the app's ease of use, quick learning curve, and responsiveness, noting that it was faster and simpler than reading news online. However, issues such as the visibility of the "Save to List" button and the lack of error messages were highlighted. Participants preferred more global news sources and clearer branding to distinguish the app. Technical issues, particularly with the translation and read-aloud features, were noted, suggesting a need for more robust solutions. Overall, the UAT feedback emphasized the importance of user-friendly design and cultural relevance, guiding future improvements for NewsMead.

5 Conclusion and Recommendations

While this research is one of the few studies in the Philippines to attempt co-designing a personalized news application for older adults, several limitations exist. The small sample size of the FGD may not necessarily reflect the preferences of older adults in digital news consumption. Future research may conduct larger FGD sessions to uncover similarities or contradictions from the findings of this study. Another limitation is the type of device and the level of connectivity of the FGD participants. These factors can dictate how older adults interact with technology artifacts, and future co-design studies may include FGDs comprising different groups (iPhone users, older adults using tablets, and older adults with limited connectivity).

This study dissected the needs and difficulties of older adults regarding smartphones and reading the news. Older adults do not actively consume news due to the prevalence of irrelevant content, which acts as a barrier and diminishes their interest in news consumption over time. They also perceive reading the news as an academic activity not suited to their lifestyle, deterring them from engaging with news content. When provided with a news application that considers their difficulties, they can enjoy news consumption and find the activity easier to navigate, ultimately fostering a renewed interest in staying informed and connected with current events.

In line with these insights, future applications geared toward older adults should prioritize user-centered design principles and accessibility features tailored to their needs. Incorporating features such as customizable font sizes, audio options for news consumption, and simplified navigation interfaces can enhance the usability and appeal of news applications for this demographic. Furthermore, ongoing user engagement and feedback mechanisms should be implemented to improve the user experience and address any emerging challenges or preferences among users.

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